

KEMI 1 FYR, Finland

WMO: 028630

Lat: 65.3853N

Lon: 24.0961E

Elev: 62

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.3	-0.3	-10.4	3.1	-5.1	-4.5	4.3	0.0	40.8	28.4	38.2	27.3	9.9	10	0.931

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	69.9	63.7	67.1	61.6	64.8	60.3	64.8	68.6	62.6	66.0	61.0	64.2	12.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	87.0	67.4	61.1	80.5	64.8	59.4	75.7	63.1	29.9	69.0	28.3	66.2	27.2	64.5	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.5	35.1	31.9	DB	-8.3	72.7	8.8	3.5	-14.6	75.2	-19.7	77.3	-24.7	79.2	-31.0	81.7	(4)
(5)			WB	-8.3	65.9	8.4	3.0	-14.4	68.0	-19.3	69.7	-24.0	71.4	-30.2	73.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	21.6	18.8	24.6	32.7	42.6	51.6	59.7	58.6	51.8	41.8	34.3	29.3	(6)
(7)		DBStd	15.22	9.58	11.29	7.66	4.65	6.18	5.07	4.79	4.26	4.27	5.41	6.51	8.24	(7)
(8)		HDD50	4744	880	872	788	518	245	40	1	1	31	255	471	642	(8)
(9)		HDD65	9476	1345	1292	1253	968	695	403	177	201	396	718	921	1107	(9)
(10)		CDD50	760	0	0	0	0	15	88	302	269	84	2	0	0	(10)
(11)		CDD65	19	0	0	0	0	0	1	14	4	0	0	0	0	(11)
(12)		CDH74	23	0	0	0	0	0	5	18	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.6	18.1	16.4	16.1	14.2	15.0	14.7	13.6	15.1	17.5	18.9	19.7	20.3	(14)
(15)	Precipitation	PrecAvg	23.7	1.9	1.6	1.3	1.3	1.7	2.0	2.6	2.5	2.2	2.4	2.1		(15)
(16)		PrecMax	32.4	3.5	3.5	2.4	3.1	3.4	3.8	5.1	5.7	4.6	4.1	4.4	4.7	(16)
(17)		PrecMin	17.8	0.5	0.2	0.4	0.3	0.4	0.3	0.4	0.5	0.6	0.2	0.7	0.7	(17)
(18)		PrecStd	4.5	0.9	0.9	0.6	0.8	0.7	1.0	1.1	1.4	1.2	1.0	1.1	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	38.4	39.0	48.4	66.4	69.8	76.6	70.5	61.2	53.3	48.1	41.5	(19)
(20)			MCWB	35.9	35.5	35.2	39.6	54.5	59.2	66.9	65.2	57.3	50.2	45.1	39.0	(20)
(21)		2%	DB	36.0	36.4	36.8	43.3	59.8	63.1	71.5	67.8	59.4	51.6	46.1	40.2	(21)
(22)			MCWB	34.8	34.0	33.9	37.6	51.2	55.6	64.9	62.4	56.0	49.2	44.1	38.1	(22)
(23)		5%	DB	34.7	34.1	35.1	39.9	54.5	60.3	68.8	65.9	58.4	50.2	44.3	39.2	(23)
(24)			MCWB	33.6	32.7	32.9	35.8	47.6	54.5	63.1	61.5	55.3	47.8	42.5	37.5	(24)
(25)		10%	DB	33.1	32.4	33.8	37.7	50.6	58.1	66.2	64.3	57.3	48.9	42.3	38.1	(25)
(26)			MCWB	32.1	31.4	32.2	34.5	45.1	53.6	61.6	60.3	54.5	46.4	40.3	36.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	36.1	36.0	41.9	56.5	60.8	68.5	68.1	58.5	51.3	46.1	40.2	(27)
(28)			MCDWB	37.3	38.1	38.0	45.6	63.0	66.6	72.8	70.2	60.5	52.4	46.9	40.8	(28)
(29)		2%	WB	35.2	34.4	34.5	38.5	52.0	58.1	66.5	63.6	57.3	49.9	44.6	38.7	(29)
(30)			MCDWB	35.8	35.9	36.1	42.6	59.5	61.7	71.0	66.3	59.0	51.4	45.9	39.6	(30)
(31)		5%	WB	33.9	33.0	33.4	36.4	48.4	56.0	64.1	62.2	56.0	48.4	42.6	37.7	(31)
(32)			MCDWB	34.6	34.0	34.7	38.9	52.7	59.2	68.1	65.1	57.9	50.0	43.9	39.0	(32)
(33)		10%	WB	32.4	31.8	32.3	35.2	46.1	54.2	62.0	60.9	55.0	46.7	40.8	36.6	(33)
(34)			MCDWB	33.1	32.6	33.5	37.2	50.4	57.3	65.6	63.6	56.9	48.5	42.1	38.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	8.3	8.4	7.1	8.5	6.6	5.8	4.6	4.3	4.7	5.5	6.1	(35)
(36)			MCDBR	5.8	6.5	6.7	9.7	15.8	10.5	8.4	5.9	4.5	4.5	5.4	5.4	(36)
(37)			MCWBR	5.7	6.0	5.6	6.2	8.5	5.8	5.4	4.8	4.3	4.3	5.7	5.1	(37)
(38)		5% WB	MCDWB	5.7	6.7	6.2	8.6	14.2	9.0	7.0	5.1	4.1	4.4	5.0	5.4	(38)
(39)			MCWBR	5.8	6.2	5.4	6.0	8.7	6.2	5.5	4.8	4.0	4.5	5.4	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.207	0.271	0.300	0.336	0.343	0.348	0.366	0.352	0.328	0.305	0.343	0.353	(40)
(41)		taud		1.943	2.075	2.116	2.143	2.400	2.499	2.468	2.498	2.538	2.353	2.559	2.595	(41)
(42)		Ebn at Noon		86	187	240	260	273	275	265	258	240	181	81	30	(42)
(43)		Edh at Noon		7	19	29	36	31	29	30	26	20	15	6	3	(43)
(44)	All-Sky Solar Radiation	RadAvg		29	191	672	1235	1606	1808	1665	1221	658	238	46	7	(44)
(45)		RadStd		3	37	66	86	142	138	144	110	74	41	8	1	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page